A COUNTEREXAMPLE TO BOX-HALF-INTEGRALITY OF THE INTERSECTION OF CROSSING SUBMODULAR FLOW SYSTEMS

MICHEL X. GOEMANS AND YUCHONG PAN

Abstract. Abdi, Cornuéjols and Zambelli proved that the intersection of two crossing submodular flow systems on a weakly connected digraph is totally dual integral, but not box-integral. Abdi conjectured that this polyhedral system is box-half-integral. We exhibit a counterexample to disprove their conjecture.

We say that a family $\mathcal{C}$ of subsets of a ground set V is *crossing* if for all $U, W \in \mathcal{C}$ with $U \cap W \neq \emptyset$ and $U \cup W \neq V$, we have $U \cap W, U \cup W \in \mathcal{C}$, and that $\mathcal{C}$ is *cross-free* if for all $U, W \in \mathcal{C}$, we have $U \subseteq W$ or $W \subseteq U$ or $U \cap W = \emptyset$ or $U \cup W = V$. We say that a set function $f : \mathcal{C} \to \mathbb{R}$ is *crossing submodular* if for all $U, W \in \mathcal{C}$ with $U \cap W \neq \emptyset$ and $U \cup W \neq V$,

$$f(U \cap W) + f(U \cup W) \leq f(U) + f(W).$$

In [1], Abdi, Cornuéjols and Zambelli proved the following theorem:

Theorem 1 (Abdi, Cornuéjols and Zambelli, 2023, [1])**.** *Let $D = (V, A)$ be a weakly connected digraph. Let $\mathcal{C}_i$ be a crossing family over V and $f_i : \mathcal{C}_i \to \mathbb{Z}$ a crossing submodular function for $i = 1, 2$. Then*

$$\begin{aligned} y\left(\delta^+(U)\right) - y\left(\delta^-(U)\right) &\leq f_1(U), && \forall U \in \mathcal{C}_1, \\ y\left(\delta^+(U)\right) - y\left(\delta^-(U)\right) &\leq f_2(U), && \forall U \in \mathcal{C}_2, \end{aligned}$$

is totally dual integral.

In addition, they gave an instance for which this system is not box-integral. In a talk given in the "Combinatorics and Optimization" workshop at ICERM in 2023, Abdi proposed the following conjecture:[1]

Conjecture 2. *Let $D = (V, A)$ be a weakly connected digraph. Let $\mathcal{C}_i$ be a crossing family over V and $f_i : \mathcal{C}_i \to \mathbb{Z}$ a crossing submodular function for $i = 1, 2$. Then*

$$\begin{aligned} y\left(\delta^+(U)\right) - y\left(\delta^-(U)\right) &\leq f_1(U), && \forall U \in \mathcal{C}_1, \\ y\left(\delta^+(U)\right) - y\left(\delta^-(U)\right) &\leq f_2(U), && \forall U \in \mathcal{C}_2, \end{aligned}$$

is box-half-integral.

---

[1] See the last slide of their talk, available at `https://app.icerm.brown.edu/assets/403/4951/4951_3700_Abdi_032820231400_Slides.pdf`.

In this short note, we disprove this conjecture, making Theorem 1 more interesting.

Theorem 3. *There exist a weakly connected digraph $D = (V, A)$, crossing submodular functions $f_i : \mathcal{C}_i \to \mathbb{Z}$ for $i = 1, 2$ and $\ell, u \in \mathbb{Z}^A$ such that*

$$y\left(\delta^+(U)\right) - y\left(\delta^-(U)\right) \leq f_1(U), \qquad \forall U \in \mathcal{C}_1, \tag{1a}$$

$$y\left(\delta^+(U)\right) - y\left(\delta^-(U)\right) \leq f_2(U), \qquad \forall U \in \mathcal{C}_2, \tag{1b}$$

$$\ell_e \leq y_e \leq u_e, \qquad \forall e \in A, \tag{1c}$$

is not half-integral.

Proof. Let $D = (V, A)$ be the weakly connected digraph in Figure 1a. Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be families of subsets of V depicted in Figures 1b and 1c, with corresponding set functions $f_i : \mathcal{C}_i \to \mathbb{Z}$ for $i = 1, 2$. Since each of $\mathcal{C}_1$ and $\mathcal{C}_2$ is a family of disjoint sets, $\mathcal{C}_1$ and $\mathcal{C}_2$ are trivially crossing families, and f_1 and f_2 are trivially crossing submodular. Let $\ell = -2 \cdot \mathbf{1} \in \mathbb{Z}^A$ and $u = \mathbf{1} \in \mathbb{Z}^A$. Let $y^* \in \mathbb{Z}^A$ be the vector defined in Figure 1d. It is easy to check that y^* is a feasible solution to the system (1).

To see that y^* is an extreme point of the polyhedron determined by the system (1), it suffices to exhibit 6 linearly independent inequalities that are tight at y^*. Since $y_e = u_e = 1$ for $e \in \{(v_1, v_2), (v_3, v_4), (v_5, v_6)\}$, we obtain 3 inequalities that are tight at y^*. In addition, it is easy to check $y(\delta^+(U)) - y(\delta^-(U)) = f_1(U) = 1$ for all $U \in \mathcal{C}_1$ and $y(\delta^+(U)) - y(\delta^-(U)) = f_2(U) = 0$ for all $U \in \mathcal{C}_2$, yielding 3 more inequalities that are tight at y^*. Moreover, it can be checked that these 6 inequalities are linearly independent (e.g., by computing the determinant of the matrix whose columns are these vectors). This gives 6 linearly independent inequalities that are tight at y^*, completing the proof. □

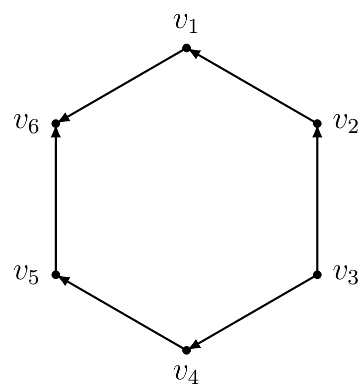
Remark. The intuition of the proof is the following. Let M_1 and M_2 be matrices with the same number of rows such that each row consists of at most one 1-entry and at most one -1-entry, with all other entries equal to 0. Then M_1 and M_2 are submatrices of incidence matrices of digraphs and are hence totally unimodular. Let

$$M = \left[\begin{array}{c|c} M_1 & M_2 \end{array}\right].$$

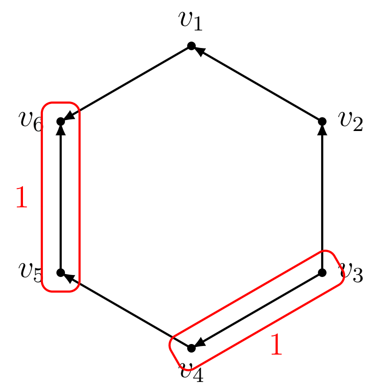
Note that M is not necessarily totally unimodular, and there exist instances such that $|\det(M)|$ can be made arbitrarily large. Take such a square matrix M with $|\det(M)| \geq 3$. Let b be an integral vector such that $(M^{\mathrm{T}})^{-1} b$ is not half-integral. In the counterexample given in the proof,

$$M = \left[\begin{array}{rr|r} 1 & 0 & -1 \\ 0 & -1 & -1 \\ 1 & -1 & 1 \end{array}\right], \qquad b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix},$$

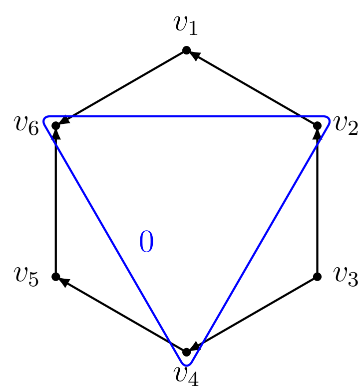
where $\det(M) = -3$.



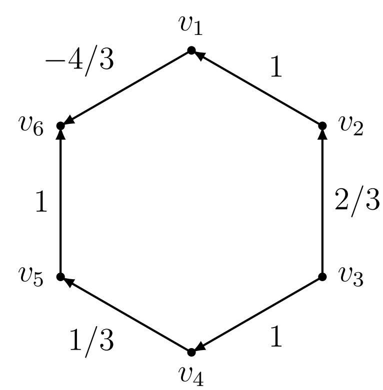
(A) The digraph $D = (V, A)$ in the proof of Theorem 3.



(B) The crossing family $\mathcal{C}_1$ and the crossing submodular set function $f_1 : \mathcal{C}_1 \to \mathbb{Z}$ in the proof of Theorem 3.



(C) The crossing family $\mathcal{C}_2$ and the crossing submodular set function $f_2 : \mathcal{C}_2 \to \mathbb{Z}$ in the proof of Theorem 3.



(D) A solution $y^* \in \mathbb{Z}^A$ to the system of linear inequalities in the proof of Theorem 3.

Let k_1 and k_2 be the numbers of columns of M from M_1 and from M_2, respectively. Let m be the number of rows of M. Let $D_0 = (V, A_0)$ be a digraph, where $V = \{0, \ldots, k_1\} \times \{0, \ldots, k_2\}$ and A_0 has m arcs, one corresponding to each row of M. For each row of M, let α, α' be the indices of the 1-entries from M_1 and from M_2 (or 0 if not present), respectively, and β, β' the indices of -1-entries from M_1 and from M_2 (or 0 if not present), respectively. Then this row corresponds to the arc $((\alpha, \alpha'), (\beta, \beta'))$.

Let $S_i = \{i\} \times \{0, \ldots, k_2\}$ for each $i \in [k_1]$. Let $T_i = \{0, \ldots, k_1\} \times \{i\}$ for each $i \in [k_2]$. Let $\mathcal{C}_1 = \{S_1, \ldots, S_{k_1}\}$ and $\mathcal{C}_2 = \{T_1, \ldots, T_{k_2}\}$. Let $g_1 : \mathcal{C}_1 \to \mathbb{Z}$ be defined by $g_1(S_i) = b_i$ for all $i \in [k_1]$, and let $g_2 : \mathcal{C}_2 \to \mathbb{Z}$ be defined by $g_2(T_i) = b_{i+k_1}$ for all $i \in [k_2]$. For $i = 1, 2$, since the sets in $\mathcal{C}_i$ are disjoint, $\mathcal{C}_i$ is trivally a crossing family and g_i is trivially crossing submodular. The system $M^{\mathrm{T}} y \le b$ is exactly

$$\begin{aligned} y\left(\delta^+(U)\right) - y\left(\delta^-(U)\right) &\le g_1(U), && \forall U \in \mathcal{C}_1, \\ y\left(\delta^+(U)\right) - y\left(\delta^-(U)\right) &\le g_2(U), && \forall U \in \mathcal{C}_2. \end{aligned}$$

Let $y_0 \in \mathbb{Q}^A$ be the unique solution to the system $M^{\mathrm{T}} y = b$.

Now, we arbitrarily add edges to D_0 to create a weakly connected digraph $D = (V, A)$. For each new arc $e \in A \setminus A_0$, add a constraint $y_e \le 1$. This results in a system $(M')^{\mathrm{T}} y \le b'$, where

$$M' = \begin{bmatrix} M & M'' \\ 0 & I \end{bmatrix}, \qquad b' = \begin{bmatrix} b + b'' \\ \mathbf{1} \end{bmatrix},$$

where each row of M'' is the vector $\gamma(U) := \delta^+_{A \setminus A_0}(U) - \delta^-_{A \setminus A_0}(U)$ for $U \in \mathcal{C}_1$ and for $U \in \mathcal{C}_2$, and where each component of b'' is $\mathbf{1}^{\mathrm{T}} \gamma(U)$ for $U \in \mathcal{C}_1$ and for $U \in \mathcal{C}_2$. By Schur's formula, $\det(M') = \det(M) \cdot \det(I) = \det(M)$. Hence, $|\det(M')| \ge 3$. Let

$$y^* = \begin{bmatrix} y_0 \\ \mathbf{1} \end{bmatrix}.$$

It is easy to check that y^* is the unique solution to the system $(M')^{\mathrm{T}} y = b$.

With a computer program and the procedure described above, one can find counterexamples to Conjecture 2 with arbitrarily large sizes.

References

[1] Ahmad Abdi, Gérard Cornuéjols, and Giacomo Zambelli. Arc connectivity and submodular flows in digraphs. `https://www.ahmadabdi.com/papers/connectedflip.pdf`.